

```
#Import Libraries
import pandas as pd
import numpy as np

df = pd.read_csv("IMDB Dataset.csv")
print(df.head())

#Display first 10 records of dataset
print(df.head(10))

#Find total number of movies/reviews in dataset
print("Total Records:", len(df))

#Display column names
print(df.columns)

#Find dataset shape (rows and columns)
print(df.shape)

#Check missing values in each column
print(df.isnull().sum())

#Display data types of all columns
print(df.dtypes)

#Count positive and negative sentiments
print(df['sentiment'].value_counts())

#Find percentage of positive reviews
positive = (df['sentiment'] == 'positive').sum()
total = len(df)
print((positive/total)*100)

#Find percentage of negative reviews
negative = (df['sentiment'] == 'negative').sum()
total = len(df)
print((negative/total)*100)

#Find longest review text length
df['review_length'] = df['review'].str.len()
print(df['review_length'].max())

#Find shortest review text length
print(df['review_length'].min())
```

```
#Find average review length
print(df['review_length'].mean())

#Display top 5 longest reviews
print(df.nlargest(5, 'review_length'))

#Count total words in each review using NumPy
df['word_count'] = np.array(df['review'].str.split().str.len())
print(df['word_count'])

#Find average word count of reviews
print(df['word_count'].mean())

#Find maximum word count review
print(df['word_count'].max())

#Find minimum word count review
print(df['word_count'].min())

#Display reviews having more than 500 words
print(df[df['word_count'] > 500])

#Sort dataset by review length descending
print(df.sort_values(by='review_length', ascending=False))

#Generate statistical summary of numeric columns
print(df.describe())

#Convert sentiment to numeric using NumPy
df['sentiment_num'] = np.where(df['sentiment']=='positive',1,0)
print(df.head())
```